Docker Machine uses the adapter pattern,[11] providing different *drivers* capable of creating instances on a wide variety of platforms.[12] We'll use this shortly to create local VirtualBox instances on page 155 and then create cloud-based infrastructure on page 173.

Sometimes you need to provision more than just instances. Perhaps you may have security constraints that need to be enforced, or maybe specific networking requirements that involve the creation of networking devices (firewalls, NAT gateways, and so on) in a specific arrangement. In this case, Docker Machine alone wouldn't be enough; you'd have to consider other tools to assist.

Chef, Puppet, and Ansible

Traditionally, configuration management tools like Chef, Ansible, and Puppet have been used to provision infrastructure and configure it, including installing and managing software on instances. However, as we've already discussed, since Docker images are now responsible for most of your server configuration, the need for these other tools is greatly diminished.

Additionally, these tools all predate Docker—especially Puppet (2005) and Chef (2009)—so they weren't born out of a worldview that included containers. That said, they can still be used to bootstrap a cluster with Docker and your orchestration layer, but depending on your needs, may be overkill.

Terraform

Released in 2014, a year *after* Docker, HashiCorp's Terraform is the relative newcomer to the scene. Rather than being a fully fledged configuration management system, it considers itself an *infrastructure orchestrator*, with the more modest aim of provisioning and updating your infrastructure in a safe, controlled way.

It has only lightweight capabilities for configuring servers, allowing you to use other tools for this where it makes sense.[13] However, because only minimal server configuration is needed when using Docker, going Terraform-only is a lightweight and popular choice.[14]

11. https://en.wikipedia.org/wiki/Adapter_pattern
12. https://docs.docker.com/machine/drivers/
13. https://www.terraform.io/intro/vs/chef-puppet.html
14. https://blog.gruntwork.io/why-we-use-terraform-and-not-chef-puppet-ansible-saltstack-or-cloudformation-7989dad2865c